

Preet Dave

AI / ML Engineer

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PROFILE

AI/ML Engineer based in Ahmedabad, India, specializing in applied machine learning, large language models, and agentic AI systems. Final-year B.Tech (CSE — AI & ML) student at NLJIET (GTU) who progressed from machine-learning intern to full-time AI/ML software engineer at a Gujarat startup, delivering ML models in direct collaboration with clients. Builds end-to-end systems spanning classical ML (scikit-learn, XGBoost), deep learning (PyTorch, TensorFlow), computer vision (YOLOv8, CLIP, TrOCR), and LLM/agent stacks (Claude API, Claude Agent SDK, Model Context Protocol, Google Gemini ADK, RAG). Hackathon builder with projects for ISRO, city police, and national competitions, plus an authored research paper and technical blogs.

EXPERIENCE

AI/ML Software Engineer · [Agility](#)

Mar 2026 — Present

Gujarat, India · On-site

- Develop machine-learning models in direct collaboration with clients, translating business requirements into model specifications and integrating them into existing software products.
- Lead delivery on applied AI/ML projects, owning problem framing, model selection, and integration to build a strong production foundation in AI and ML.
- Deliver hands-on, real-world industry work at a dynamic Gujarat startup, balancing rapid iteration with reliable client outcomes.
- Promoted from machine-learning intern to full-time engineer after demonstrating delivery across a 10-month total tenure.

Machine Learning Intern · [Agility](#)

Sep 2025 — Mar 2026

Gujarat, India · On-site

- Collaborated on ML projects that enhanced client automation processes, contributing to data preparation, modeling, and integration.
- Engaged directly with clients to understand needs and help deliver tailored ML solutions.
- Assisted in deploying predictive models that improved operational efficiency for client workflows, earning a full-time offer.

Project Lead · [Gujarat Technological University \(GTU\)](#)

Oct 2023 — Present

Gujarat, India · On-site

- Lead development of CodeTransform, a Python ↔ JavaScript code converter, from inception through a state-level Design Engineering presentation.
- Designed the application architecture while concurrently building the product, balancing technical design with hands-on implementation.
- Coordinated cross-functional student teams to align goals, deliverables, and timelines; drove the project from prototype to a demonstrable web app with auto language detection and confidence-scored conversions.

Remote

- Took ownership of ML models and pipelines, applying reinforcement-learning techniques for the product's core logic.
- Built an MVP end-to-end and pitched it to potential business partners; helped prepare the project for a Y Combinator application.
- Operated with high autonomy in a small remote team, contributing across modeling, pipeline engineering, and product framing.

Team Lead · [Smart India Hackathon \(SIH\)](#)

Aug 2024 — Dec 2024

India · National

- Directed a team building a virtual herbal garden featuring detailed interactive 3D models, presented at the national level.
- Managed team dynamics, task allocation, and coordination for smooth execution under competition deadlines.

PROJECTS**Therma Netra — Digital Twin of India's Climate · [CLIMATE ML](#) · [DIGITAL TWIN](#)**

An AI-powered digital twin of India's climate built entirely on national datasets (IMD, INSAT/MOSDAC, IMDAA, CPCB). One continuously-assimilated twin powers climate state & forecast, urban-heat mitigation, and surface air quality, achieving ~30% forecast skill over a persistence-of-anomaly baseline. Built for ISRO's Bharatiya Antariksh Hackathon (BAH) 2026, Problem Statement #5.

Python · Streamlit · xarray · imdlib · Pydeck · Climate ML · Digital Twin

CityShield · VisionScan — Unified AI Policing Platform · [COMPUTER VISION](#) · [GOVTECH](#)

A unified AI policing platform fusing physical and cyber crime on one GIS map, forecasting next-week risk, and closing the loop from a live CCTV anomaly to a dispatched patrol unit — fully offline on CPU. Four CCTV search modes (CLIP→FAISS, ArcFace, YOLOv8), 7 generated statutory documents (en/hi/gu), a 4-layer OWASP middleware, and 77 passing backend tests. Risk model backtested on synthetic data (Hit-Rate@10 0.771, PAI@10 2.31x over 8 weekly folds) as a methodology demonstration. Built for KANAD S.H.I.E.L.D. 2026 with the Cyber Crime Branch, Ahmedabad City Police. github.com/iampreetdave-max/Guardian-Angle

FastAPI · React · CLIP · FAISS · YOLOv8 · InsightFace · RAG · Docker

Arbiter — Agentic Legal AI · [AGENTIC AI](#) · [LEGAL TECH](#)

A full-stack AI legal assistant that turns a problem described in plain English or Hindi into a structured legal document with no upfront lawyer fees. Gemini agents (Google ADK) research applicable law in real time and draft with citations across 6 document types and 5 jurisdictions. Deployed on Google Cloud Run (asia-south1). Built for XPRIZE "Build with Gemini" 2026. github.com/iampreetdave-max/arbiter

Next.js 14 · FastAPI · Google Gemini 2.0 · Google Agent Development Kit · Firebase · Cloud Run · Razorpay

AI Race News — AI/ML News Aggregation Platform · [DATA ENGINEERING](#) · [BACKEND](#)

A production platform that ingests 110+ AI/ML sources every 15 minutes via APScheduler, deduplicates with a 3-layer pipeline (URL, content hash, title similarity), auto-tags across 12 categories, and serves 5 audience-specific feeds through a public FastAPI REST API and a Next.js frontend. github.com/iampreetdave-max/ai-race-news

FastAPI · Python · BeautifulSoup · APScheduler · Docker · Next.js · SQLite · pytest

Sports Match-Prediction Engines · [ML](#) · [SPORTS ANALYTICS](#)

A production multi-sport ML suite (NBA, soccer, NASCAR) that forecasts outcomes from engineered pre-match features and live sportsbook odds, logging every prediction to PostgreSQL and grading it against the real settled result with full P&L tracking; engines run autonomously on daily GitHub Actions workflows. Tracked results: +16.5% ROI on Grade-A NBA picks and a 66.3% moneyline win rate over 1,064 settled games; +9.1% ROI on Grade A+B soccer picks (15+ leagues, 3,020 matches); 7,566 NASCAR records; 4,084+ live predictions logged over 15+ months. Models span Ridge regression to XGBoost / LightGBM / Random Forest with Platt-scaled probabilities. github.com/iampreetdave-max/football-predictions

XGBoost · LightGBM · Ridge Regression · Python · PostgreSQL · GitHub Actions · Streamlit

CodeTransform — Python ↔ JavaScript Converter · DEVELOPER TOOLS

A web-based code converter between Python and JavaScript built on 122+ tested conversion rules, with auto language detection and color-coded confidence scoring. Zero-build vanilla-JS + Tailwind frontend on a FastAPI backend. GTU state-level Design Engineering project (Project Lead). github.com/iampreetdave-max/code-converter

FastAPI · Python · JavaScript · Tailwind CSS · Language Detection

TalkToNotes — Handwriting to Knowledge Base · NLP · DOCUMENT INTELLIGENCE

A hackathon project that digitizes handwritten and printed notes with TrOCR, then turns them into a searchable knowledge base powering a chatbot and one-click quiz generation. Built for the Rotaract Club Hackathon.

TrOCR · Transformers · Vector Search · Python · NLP · OCR

Find-Ranks — PDF Marksheet Analytics · UTILITY

A Streamlit utility that extracts marks from multiple PDF mark-sheets, computes cumulative performance, and generates rankings with analytics for educational institutions. github.com/iampreetdave-max/Find-Ranks

Streamlit · Python · PDF Processing · pandas

SKILLS

Languages: Python (advanced), C / C++ (advanced), JavaScript, TypeScript, SQL, HTML/CSS, PHP

ML & Deep Learning: scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Keras, pandas, NumPy, feature engineering, ensemble methods, regression & classification

AI / LLM & Agents: LLMs, RAG, Claude API, Claude Agent SDK, Model Context Protocol (MCP), Google Gemini ADK, prompt engineering, vector search & embeddings (FAISS, ChromaDB), NLP

Computer Vision: YOLOv8, CLIP, TrOCR, InsightFace / ArcFace, OCR

Backend & Full-stack: FastAPI, Next.js, React, Node.js, REST APIs, Streamlit

Data & Infrastructure: PostgreSQL / Supabase, SQLite, Docker, Docker Compose, GitHub Actions / CI, Azure, Google Cloud Run, Netlify, Hugging Face

Automation: Make.com, n8n, Zapier, GoHighLevel, webhook & API integrations

Practice: MLOps, model deployment, backtesting & validation, Git

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML)

2022 — 2026 (expected)

NLJIET — Gujarat Technological University (GTU), Ahmedabad, India

- Coursework & focus: Large Language Models (LLMs), Data Structures & Algorithms, AI/ML, Mathematics for ML.
- Coordinator for multiple student-council events — end-to-end planning, execution, and marketing.
- Member of the college research group; co-authoring research papers with classmates.

CERTIFICATIONS

Building with the Claude API — Anthropic, 2026 (ID tgbfokqc8ome)

Claude Code in Action — Anthropic, 2026 (ID 2zrdb9sy8hf)

Introduction to Model Context Protocol — Anthropic, 2026 (ID 56bth895hwgs)

Introduction to Agent Skills — Anthropic, 2026 (ID 6xts2pa7yvoy)

Data Literacy — IBM, 2026

Crash Course on Python — Google, 2023 (ID JH9QK9VM8QM8)

Make Automation Certification — Make, 2026

PUBLICATIONS

Engineers Fear AI, As Mathematicians Once Feared Calculators — authored research paper

Zipf's Law for LLMs — technical essay (LinkedIn)

How GPS Works — technical essay (LinkedIn)

LEADERSHIP & ACHIEVEMENTS

- Promoted from Machine Learning Intern to full-time AI/ML Software Engineer at Agility within a 10-month tenure.
- Project Lead at GTU: drove the CodeTransform converter from inception to a state-level Design Engineering presentation.
- Team Lead, Smart India Hackathon 2024: led a team to the national level with a virtual herbal garden featuring 3D models.
- Built CityShield · VisionScan for KANAD S.H.I.E.L.D. 2026 with the Cyber Crime Branch, Ahmedabad City Police.
- Built Therma Netra for ISRO's Bharatiya Antariksh Hackathon (BAH) 2026 (Problem Statement #5).
- Coordinator for multiple GTU student-council events; member of the college research group.